

AI-ITI Study Assistant

RAG-Powered Document Assistant over course lab material

Ali Ezz Ali · Level 2 Summer Training · Core Track · Individual

github.com/ali-ezz/ai-iti-rag-assistant

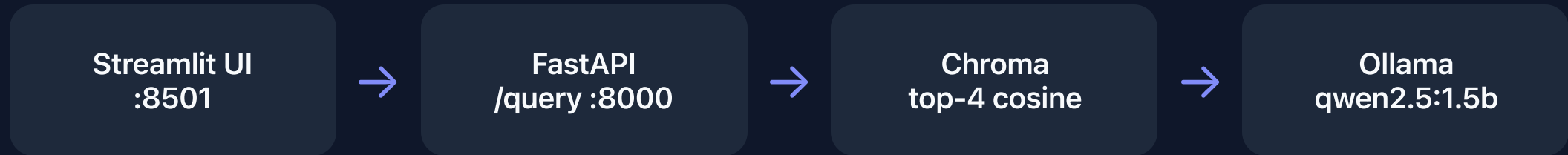
Problem & Goal

Grounded answers only from course material

- ▶ Students need quick, trustworthy answers from 12 lab notebooks (OOP → Transformers).
- ▶ A plain LLM invents content — unacceptable for study help.
- ▶ Goal: full product — raw docs → notebook → FastAPI → Streamlit → GitHub.
- ▶ Answer must cite which document / section / chunk it came from.

Architecture

Question → retrieve → grounded generate → cited answer



Embeddings: paraphrase-multilingual-MiniLM-L12-v2 · persisted store · temp 0.1 · [Source N] citations

Data & Chunking

Core Track — text-only corpus

- ▶ 12 solved lab notebooks → cleaned Markdown — no OCR needed.
- ▶ Split by Markdown section, then 320-word windows with 60-word overlap.
- ▶ 264 chunks indexed into Chroma (cosine HNSW) — persisted for the API.
- ▶ Metadata on every chunk: source, section, chunk_id → human-readable citations.

Grounding & anti-hallucination

Strict context-only generation

- ▶ System prompt: answer ONLY from supplied context; refusal if insufficient.
- ▶ Numbered [Source 1..k] context blocks; model must cite factual claims.
- ▶ Low temperature (0.1), short max tokens, de-duplicated sources in API response.
- ▶ UI shows expandable source cards under every answer.

Evaluation

12 test questions · CI-run notebook · results.csv

12 / 12

Retrieval hit rate @4

12 / 12

Answers with citations

264

Chunks from 12 docs

- Failures: overlapping terms, broad questions, small LLM truncation.
- Mitigations: section chunks + overlap, top-4, strict prompt, refusal path.

Live demo

What we will show

- ▶ 1. Open Streamlit chat UI at localhost:8501.
- ▶ 2. Ask: "How does Non-Max Suppression remove duplicate detections?"
- ▶ 3. Show grounded answer + Sources expander (LAB-05).
- ▶ 4. Ask: stemming vs lemmatization → LAB-08.
- ▶ 5. Show GET /health → vector store + Ollama ready.
- ▶ 6. Repo: structure, tests, CI green.

Tech stack & deliverables

Stack

- Python 3.12 · FastAPI · Streamlit
- ChromaDB · sentence-transformers
- Ollama qwen2.5:1.5b · pytest
- Docker Compose · GitHub Actions
- uv · Makefile

Checklist

- ✓ Executed rag_pipeline.ipynb
- ✓ /health + /query + tests
- ✓ Chat UI with cited sources
- ✓ Persisted vector store
- ✓ README + public GitHub + 100% eval

Thank you

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Ali Ezz Ali · Level 2 · Core Track · Ready for Q&A